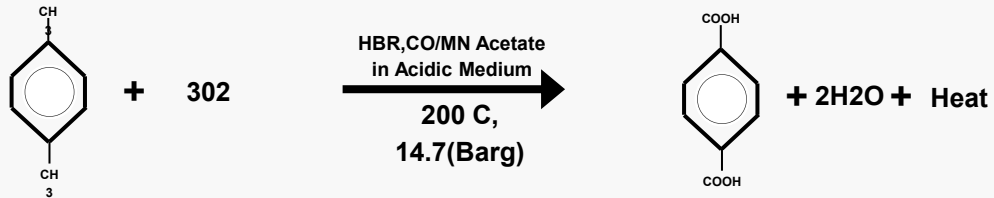


COMPRESSOR-PERFORMANCE-ANALYZER

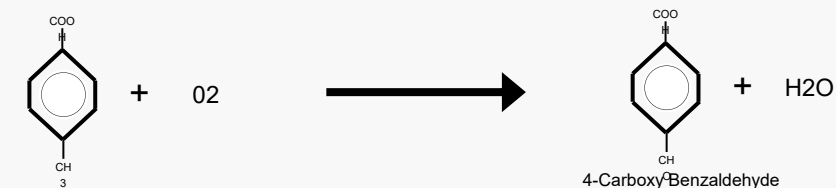
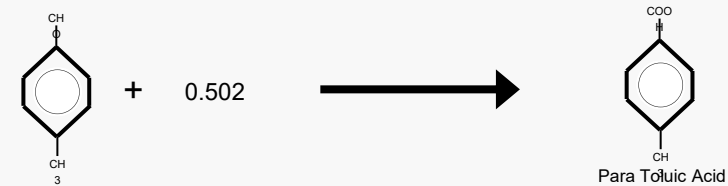
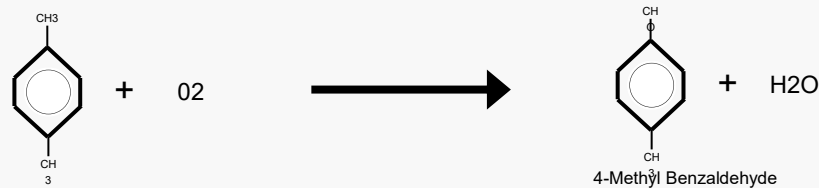
The Compressor Performance Analyzer is a sophisticated, browser-based thermodynamic simulation platform designed to help engineers, researchers, and students accurately analyze, optimize, and forecast the performance of multi-stage centrifugal air compressor systems.

Reaction Pathway for PTA Production From Para-Xylene

Overall Reaction:



Step wise Reaction :



Process Air Compressor (PAC) - Process Description

Configuration of PAC

- Five-stage centrifugal compressor.
- **Intercooler between 1st and 2nd stage.**
- **Intercooler between 2nd and 3rd stage.**
- **Intercooler between 3rd and 4th stage.**
- **Steam Turbine, Process Air Compressor, and Expander are mounted on a common shaft.**
- Inlet Guide Vanes (IGV) provided for flow control.

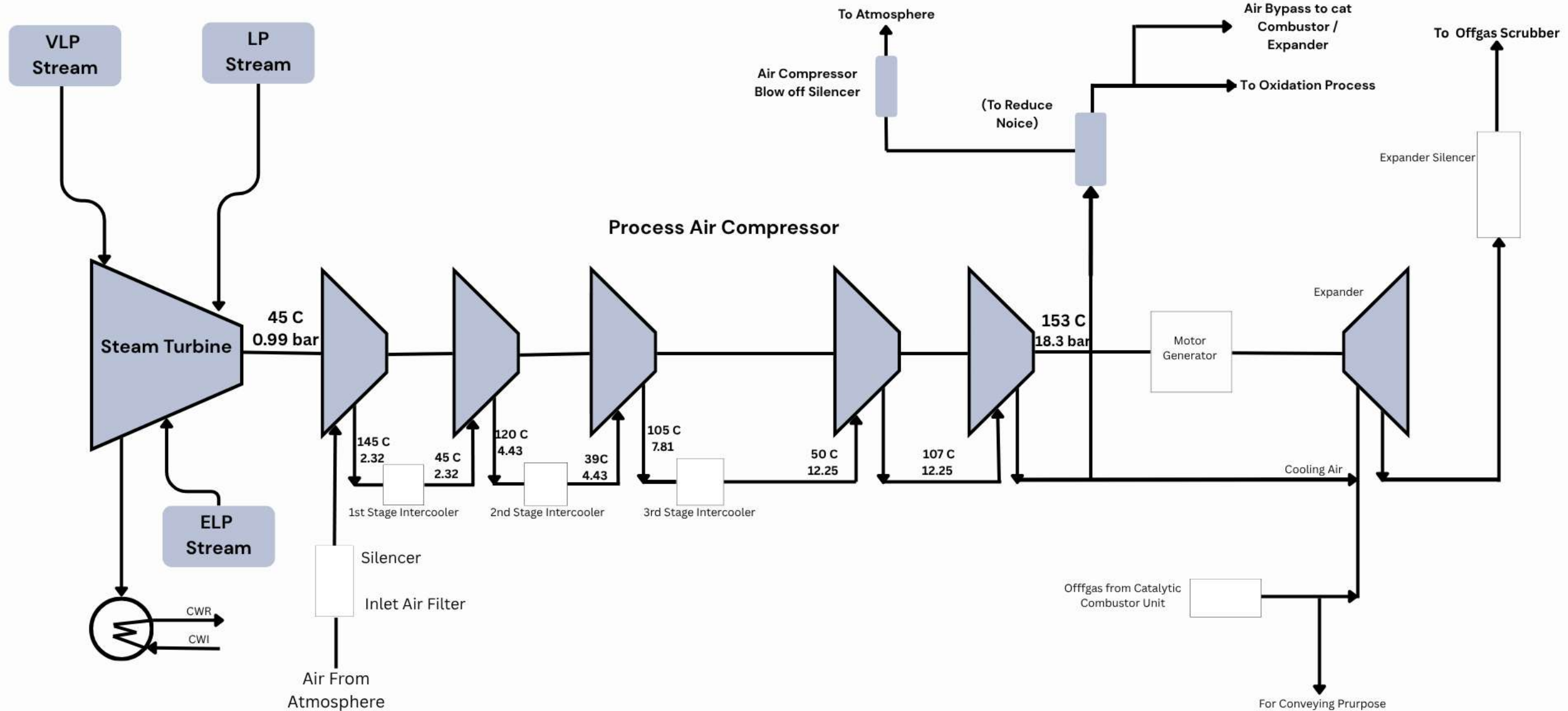
Working Principle

- Atmospheric air enters the compressor through the inlet section.
- **Air is compressed in five stages.**
- The pressure increases progressively in each stage.
- **Compression increases air temperature.**
- Intercoolers remove heat between stages.
- **Compressed air leaves the fifth stage at the required process pressure and temperature.**
- The compressed air is then supplied to downstream process units.

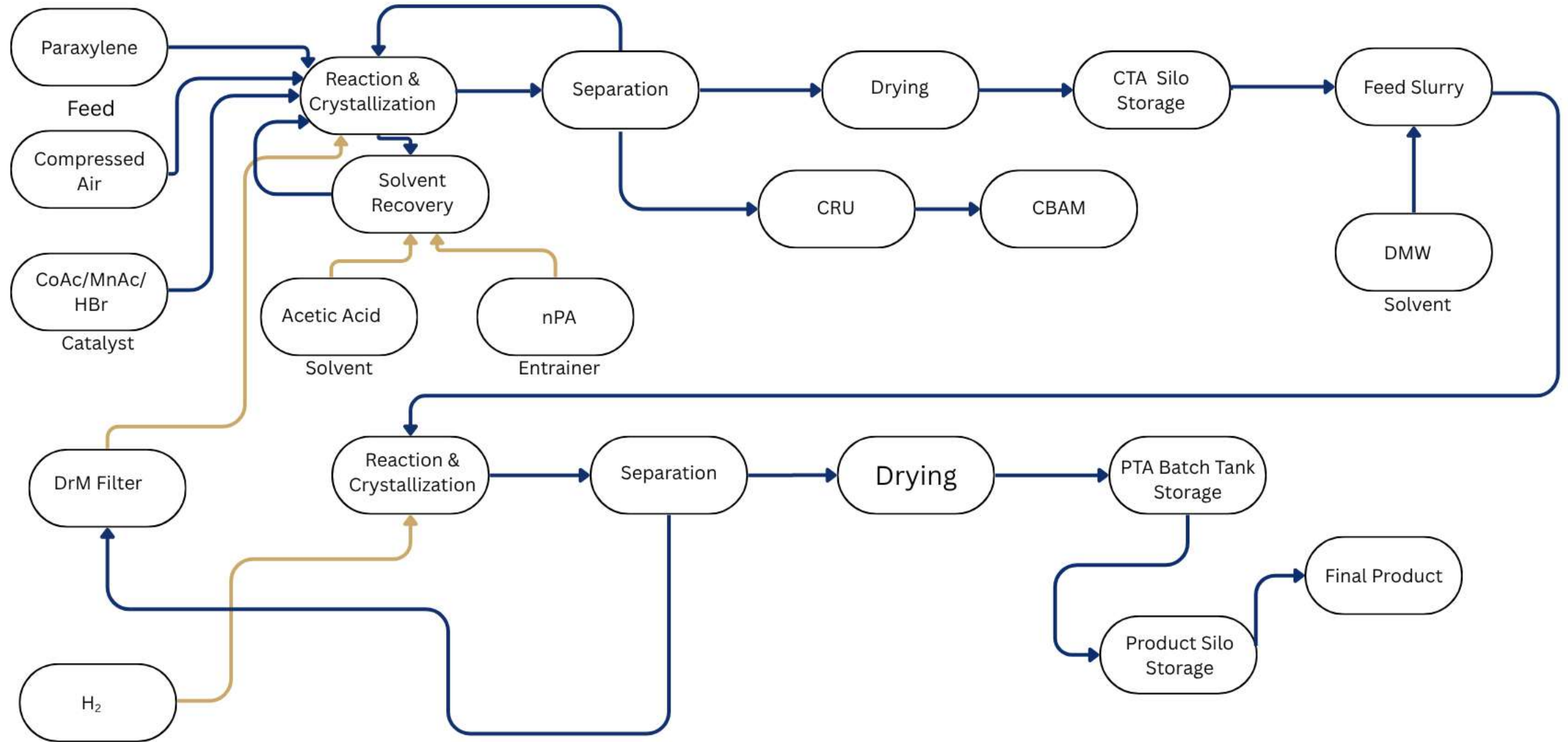
Compressor Drive System

- **During start-up, the compressor is driven by an electric motor.**
- During normal operation, the compressor is driven by a steam turbine and expander.
- **Integrated turbine-expander arrangement enables energy recovery and utilization.**
- This arrangement improves plant energy efficiency.

Process Air Compressor: An Overview



Process Flow Diagram-PTA



App Screenshot

Input Parameters

STEP 1

Number of Stages

5

1-20 stages

Relative Humidity *

52.2

0-100%

Atmospheric Pressure *

0.9738

bar

Atmospheric Temperature *

27.3

°C

Air Flow *

1771

kg/hr

Specific Heat Ratio (k)

1.4

Dimensionless

Gas Constant

0.287

kJ/kgK

Intercooler Efficiency

0.84

0-1

Compression Type

Polytropic

Select compression model

Mechanical Efficiency

0.95

0-1

Motor Efficiency

0.92

0-1

Intercooler Pressure Loss

0.022

bar

Energy & Cost Parameters

Annual Operating Hours

8000

hours/year

Electricity Tariff

7.5

₹/kWh

Currency

₹ (INR)

Select currency

Generate Stages

Auto-Fill

AI Optimization & Analysis

ADVANCED

Current Annual Energy Cost: ₹ 14,628,745.144

● Increase Intercooler Efficiency

Current: 84% → Recommended: 95%

💡 Potential Savings: 24137

Comparison of Design and Calculated Inlet Air Conditions :

Design

Relative Humidity	60	%
Atmospheric Pressure	0.99	bar
Suction Pressure	99000	Pa
Temperaturein	45	C
Vapour Pressure of Water Vapour	9552	Pa
Partial Pressure of Water Vapour	5731	Pa
Mole Fraction(Xw) of water in	0.05789	

Calculated

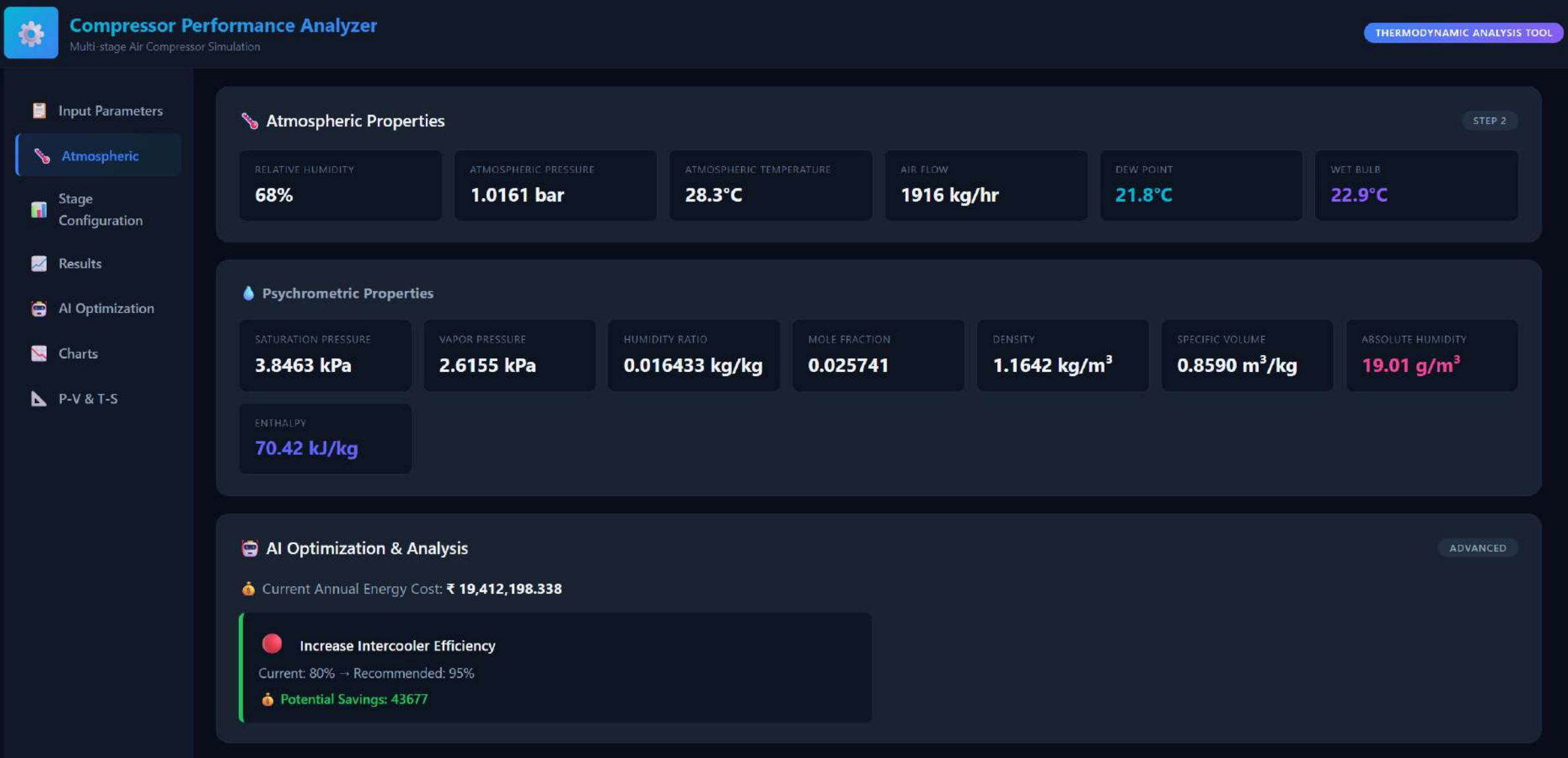
Relative Humidity	65	%
Atmospheric Pressure	0.9932	bar
Suction Pressure	99320	Pa
Temperaturein	34	C
Vapour Pressure of Water Vapour	5302	Pa
Partial Pressure of Water Vapour	3446	Pa
Mole Fraction(Xw) of water in	0.03470	

Stage-wise Water Vapour Mole Fraction Analysis (Design Case) :

Design

Stage	Xwin	Pin	Tin	Pstage out	Tstage out	Pcoolerstag e,out	Tcoolerst age,out	V.Pstage out	Xwout	Remarks
1	0.05789	0.99	45	2.32	145	2.319	40	7352.789	0.03171	Condensation
2	0.03171	2.319	40	4.43	120	4.43	39	6969.77	.01573	Condensation
3	0.01573	4.43	39	8.19	105	7.81	50	12296.2	.01501	Condensation
4	0.01501	8.19	50	12.25	107	12.25	107	129570.3	.01501	No Condensation
5	0.01501	13.05	107	18.3	153	18.3	153	511403	.01501	No Condensation

App Screenshot

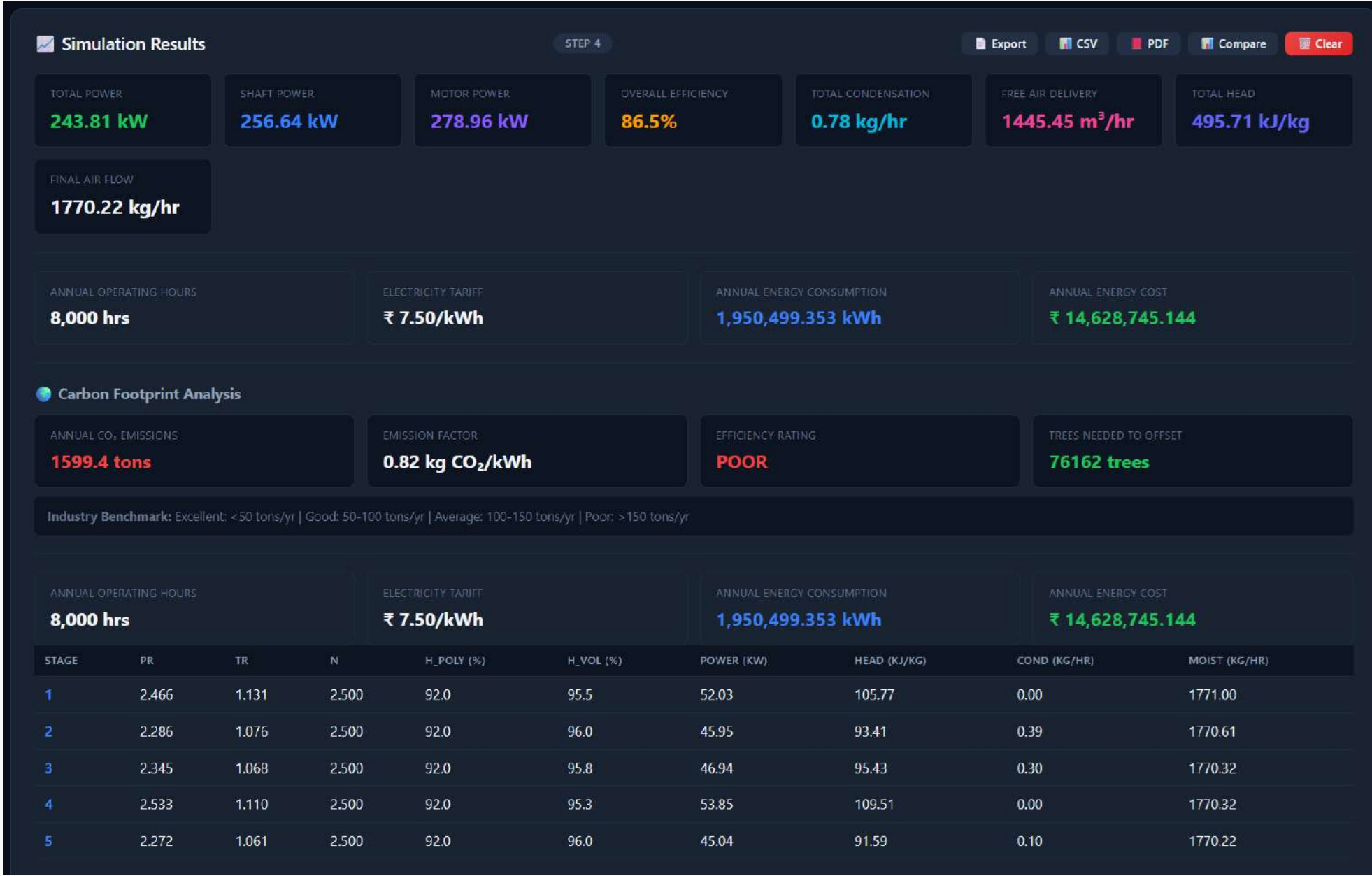


Stage-wise Water Vapour Mole Fraction Analysis (Calculated Case) :

Calculated

Stage	Xw,in	P,in	T,in	Pstageout	Tstageout	Pcoolers tage,out	Tcoolers tage,out	VP stage out	Xw,out	Remarks
1	0.03470	0.9932	34	2.34	138.3	2.3432	55.8	16301.2	0.03470	No Condensation
2	0.03470	2.3432	55.8	4.44	140.8	4.4432	39.9	7313.687	0.01646	Condensation
3	0.01646	4.4432	39.9	7.3532	107.9	7.81	43.8	8978.649	0.01221	Condensation
4	0.01221	7.3532	43.8	12.3932	106.8	12.3932	106.8	128698.1	0.01221	No Condensation
5	0.01221	13.05	106.8	18.3	153	18.3	153	511403.4	.01221	No Condensation

App Screenshot



Design Case - Stage-wise Water Condensation :

Design

Stage	Xw,in	Xw,out	Remarks	Water In (kmol/hr)	Water Out (kmol/hr)	Water Condensed (kg/hr)
1	0.05789	0.03171	Condensation	1039.725	554.026	8742.6
2	0.03171	.01573	Condensation	554.026	270.451	5104.4
3	0.01573	.01501	Condensation	270.451	257.895	226.0
4	0.01501	.01501	No Condensation	257.895	257.895	0.0
5	0.01501	.01501	No Condensation	257.895	257.895	0.0

Comparison of Mass of Water in Air at Suction Condition (Design vs. Calculated) :

Design

Mw of water from 5th stage discharge	28.80264542	g/mol	--	--
Moles of Wet Air	17959.16229	Kmole/ hr	510.0402	t/hr
moles of dry air	16919.43686	kmole/ hr	480.512	t/hr
Mass of water in air at suction condition	---	---	29.5282	t/hr

Calculated

Mw of water from 5th stage discharge	28.8333313	g/mol	--	--
Moles of Wet Air	18662.46449	Kmole/ hr	530.014	t/hr
moles of dry air	18014.88983	kmole/ hr	511.6229	t/hr
Mass of water in air at suction condition	---	---	18.39113	t/hr

Calculated Case - Stage-wise Water Condensation :

Calculated

Stage	Xw,in	Xw,out	Remarks	Water In (kmol/hr)	Water Out (kmol/hr)	Water Condensed (kg/hr)
1	0.03470	0.03470	No Condensation	647.575	647.575	0.0
2	0.03470	0.01646	Condensation	647.575	301.495	6229.4
3	0.01646	0.01221	Condensation	301.495	222.691	1418.5
4	0.01221	0.01221	No Condensation	222.691	222.691	0.0
5	0.01221	.01221	No Condensation	222.691	222.691	0.0

Comparison of Stage-wise Water Condensation Across Intercoolers (Design vs. Calculated) :

Design

Result Of all 5 stages :

Condensation in 1st stage(E-116)	8.75	t/hr
Condensation in 2nd stage(E-117)	5.11	t/hr
Condensation in 3rd stage(E-118)	0.23	t/hr
Total Water Condensed	14.0886	t/hr
Remaining Water in air	15.4	t/hr
Total Remaining air Flow	496.0	t/hr

Calculated

Result Of all 5 stages :

Condensation in 1st stage(E-116)	0.00	t/hr
Condensation in 2nd stage(E-117)	6.24	t/hr
Condensation in 3rd stage(E-118)	1.42	t/hr
Total Water Condensed	7.6564	t/hr
Remaining Water in air	10.7	t/hr
Total Remaining air Flow	522.4	t/hr

Stage-wise Compressor Efficiency Analysis (Design Case) :

Design

Result Of all 5 stages :

Stage	Efficiency(%)	Kavg.	Pout/Pin	(K-1)/K	Tout/Tin	Zout/Zin
1	88.1	1.396	2.342	0.284	1.314	0.99685
2	80.5	1.39935	1.910	0.285	1.256	1.00190
3	84.4	1.4051	1.763	0.288	1.212	1.00190
4	82.0	1.4103	1.593	0.291	1.176	1.00280
5	83.8	1.4090	1.402	0.290	1.121	1.00290

Stage-wise Compressor Efficiency Analysis (Calculated Case) :

Calculated

Result Of all 5 stages :

Stage	Kavg.	Pout/Pin	(K-1)/K	Tout/Tin	Zout/Zin	Efficiency(%)
1	1.396	2.359	0.284	1.340	0.99685	84.2
2	1.39935	1.896	0.285	1.256	1.00190	78.8
3	1.4051	1.758	0.288	1.217	1.00190	81.9
4	1.4103	1.685	0.291	1.199	1.00280	82.5
5	1.4090	1.402	0.290	1.22	1.00290	83.4

App Screenshot



App Screenshot



Findings :

Stage	Efficiency Observation	Result	Conclusion
1	Design: 88.1% Actual: 84.2%	3.9% efficiency reduction.	Highest performance deterioration; priority for improvement.
2	Design: 80.5% Actual: 78.8%	1.7% efficiency reduction.	Indicates higher thermodynamic losses and requires performance optimization.
3	Design: 84.4% Actual: 81.9%	2.5% efficiency reduction.	Moderate performance deterioration.
4	Design: 82.0% Actual: 82.5%	0.5% efficiency improvement	Operating close to design conditions.
5	Design: 83.8% Actual: 83.4%	0.4% efficiency reduction.	Stable compressor performance.
Overall PAC	Design: 83.76% Actual: 82.16%	1.6% overall efficiency reduction.	Performance degradation is mainly concentrated in Stages 1–3 .

Stage-wise Polytropic and Actual Power Analysis (Design Case) :

Design

Result Of all 5 stages :

Stage	Actual Power(MW)	Polytropic Power(KW)	Polytropic exponent, n	$n/(n-1)$	Mw inlet	Zin	Tin(K)	AirFlow(Kg/s)
1	14.43	12891	1.4665	3.149	28.33	0.9996	318.15	141.34
2	11.42	9188	1.549	2.820	28.62	0.9993	313.15	138.91
3	9.18	7744	1.519	2.926	28.79	0.9988	312.15	137.49
4	7.91	6484	1.550	2.818	28.80	0.9984	323.15	137.43
5	6.47	5416	1.530	2.886	28.80	1.0012	380.15	137.43

Stage-wise Polytropic and Actual Power Analysis (Calculated Case) :

Calculated

Result Of all 5 stages :

Stage	Polytropic exponent, n	$n/(n-1)$	Mw inlet	Zin	Tin(K)	AirFlow(Kg/s)	Polytropic Power(KW)	Actual Power(MW)
1	1.508	2.967	28.59	0.9996	307.15	148.20	13172	15.65
2	1.568	2.760	28.59	0.9993	328.95	148.20	10202	12.95
3	1.543	2.841	28.79	0.9988	313.05	146.46	8252	10.08
4	1.545	2.834	28.83	0.9984	316.95	146.07	7639	9.26
5	1.534	2.873	28.83	1.0012	379.95	146.07	5749	6.89

Result of all stage water condensation :

Design

Result Of all 5 stages :

Water condensed in 1st stage	8742.6	kg/hr
Water condensed in 2nd stage	5104.4	kg/hr
Water condensed in 3rd stage	226.00	kg/hr
Water condensed in 4th stage	0.0	kg/hr
Water condensed in 5th stage	0.0	kg/hr

Calculated

Result Of all 5 stages :

Water condensed in 1st stage	0.0	kg/hr
Water condensed in 2nd stage	6229.4	kg/hr
Water condensed in 3rd stage	1418.5	kg/hr
Water condensed in 4th stage	0.0	kg/hr
Water condensed in 5th stage	0.0	kg/hr

Future Plans

- **Shell-Side and Tube-Side Heat Balance Evaluation of intercoller.**
- Determination of the Polytropic Exponent of the PAC Compressor in PTA-5/6.
- **Polytropic Power Calculation of the PAC Compressor in PTA-5/6.**
- **Actual Power Consumption Analysis of the PAC Compressor in PTA-5/6.**
- Find which factors affecting PAC compressor efficiency and how we can resolve it.
- **What are the major causes of efficiency losses in the PAC system.**
- Impact of suction conditions, cooling efficiency, and pressure drops on overall efficiency as well as Weather.
- **Recommendations for improving PAC compressor performance and operational reliability.**

Recommendation

- **Ensure proper installation, alignment, and handling of air seal strips during maintenance and shutdown** activities to prevent damage and **air bypassing**.
- Carry out **periodic cleaning and inspection of finned tube bundles** to maintain effective heat transfer performance.
- Monitor **intercooler outlet temperature, pressure drop, and condensate** generation regularly to detect performance deterioration at an early stage.
- Maintain adequate **cooling water flow** and quality to **minimize fouling** and sustain intercooler effectiveness.
- Conduct periodic **compressor performance audits** to identify **efficiency losses** and implement corrective actions proactively.
- Adopt a **condition-based maintenance approach for intercoolers and compressor internals** to prevent unexpected **failures** and **efficiency degradation**.